AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2018

Biology Paper II

Time: 2 hours 20 minutes Marks: 55

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 10 and 11 offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. A student is interested in learning about carbon cycle, conservation of endangered species and recycling of waste. Which field of biology should he specialise in and why? (2 Marks)

_____

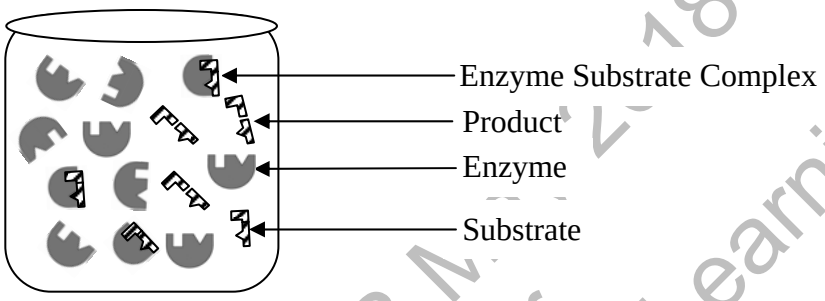
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b. The given illustration shows an enzyme catalysed reaction.



Plot a graph to show the effect of enzyme concentration on the rate of reaction for the reaction in the given illustration**.** (2 Marks)

Space for graph

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Q.2. (Total 4 Marks)

a. On the basis of five-kingdom classification system proposed by Robert Whittaker, name the kingdom which includes organisms with

i. absorptive mode of nutrition. (1 Mark)

_____

ii. ingestive mode of nutrition. (1 Mark)

_____

b. Zara cultured bacteria and viruses on artificial media containing nutrients in two different petri dishes while keeping the conditions similar. As a result bacteria successfully cultured in the artificial medium but viruses did not.

Describe the reason for the unsuccessful growth of viruses on the artificial medium. (2 Marks)

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Q.3. (Total 4 Marks)

Farah usually gets severe cold in winters. She often takes antibiotics to treat the cold without consulting a doctor.

a. Would antibiotics be effective to cure the cold? Give reason for the answer. (2 Marks)

b. Mention any TWO possible effects of the use of antibiotics by Farah without consulting a doctor. (2 Marks)

Q.4. (Total 4 Marks)

State any FOUR features of plant-like protists.

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Q.5. (Total 4 Marks)

The given diagrams show key reproductive features of two phyla of fungi.



Diagram I



Diagram II

Identify the phyla of fungi to which each of the given reproductive features belong. Give ONE reason for each identification.

Identification of diagram I: _____

Reason for identification: _____

_____

_____

Identification of diagram II: _____

Reason for identification: _____

_____

_____

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Q.6. (Total 5 Marks)

a. Describe any THREE features important for the seed to adapt to dry land. (3 Marks)

b. State any TWO outcomes of shuffling of genes in meiotic division during alternation of generation of bryophytes. (2 Marks)

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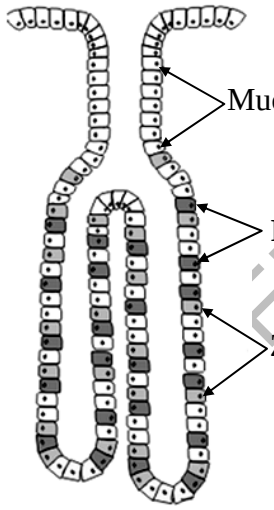
Q.7. (Total 5 Marks)

a. The given reaction shows photolysis.

$2H_2O \longrightarrow 4H^+ + 4e^- + O_2$

What is the fate of hydrogen ions (H^+) and electrons (e^-) produced in light dependent reactions of photosynthesis? (2 Marks)

b. The given diagram shows gastric glands in the stomach wall of a human being.



Identify the cells labelled as **F**. How do these cells help in the process of digestion? (3 Marks)

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Q.8. (Total 5 Marks)

A sparrow (bird) can be active at an altitude of 6000 m, whereas a mouse (mammal), which has similar body mass and metabolic rate, would die at the same altitude because of lack of oxygen.

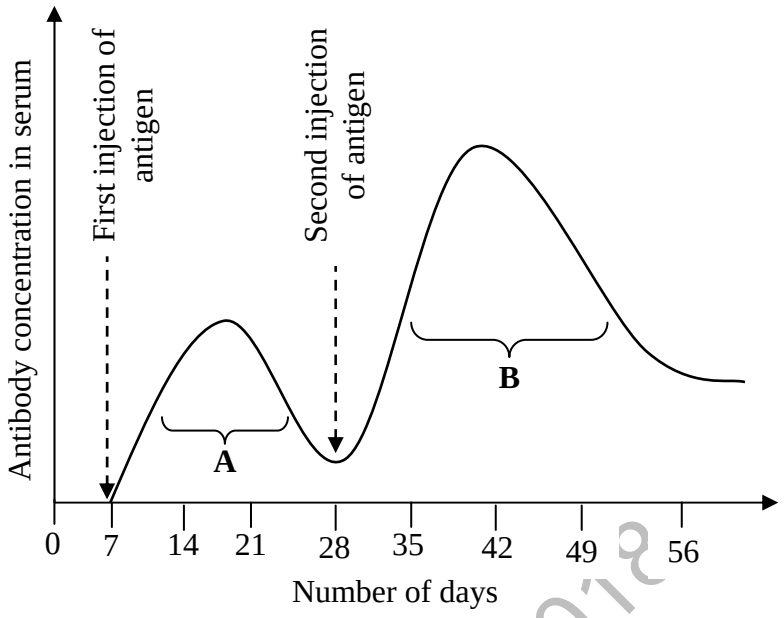
How is the respiratory system of a sparrow different from that of a mouse which enables it to be active at 6000 m?

Q.9. (Total 5 Marks)

a. A doctor listening to the heart beat through a stethoscope hears two sounds, *lub* and *dub*. How are these two sounds produced? (2 Marks)

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b. The given graph represents the profile of antibody production on the first injection of antigen and second injection of the same antigen.



i. Which type of adaptive immune response is triggered by the first injection of antigen? (1 Mark)

ii. How does the immune response **A**, produced as a result of first injection of antigen, trigger the immune response **B** produced after the second injection of the same antigen? (1 Mark)

iii. Which immune response, **A** or **B**, is rapid enough to recover a recurrent infection? (1 Mark)

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Q.10 (Total 7 Marks)

EITHER

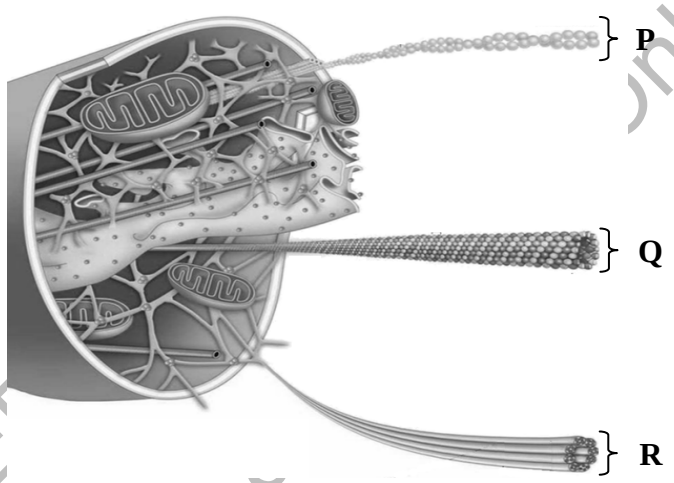
a. Lipids are the heterogeneous group of compounds.

i. Mention any ONE common property of all groups of lipids. (1 Mark)

ii. Name any THREE groups of lipids and describe the chemical structure of EACH group. (6 Marks)

OR

b. The given diagram depicts three components of cytoskeleton which provide mechanical support to the cell.



i. Describe the structure and function of the labelled components, **P**, **Q** and **R**, in the given diagram of cytoskeleton. (6 Marks)

ii. Name the protein subunits present in the labelled component **Q**. (1 Mark)

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Q.11. (Total 8 Marks)

EITHER

a. Describe the following.

i. Evolutionary origin of class Mammalia particularly exhibited by sub-class Prototheria. (2 Marks)

ii. Characteristics of sub-classes of Mammalia to which spiny ant-eater, kangaroo and whale belong. (6 Marks)

OR

b.

i. Describe the formation of acetyl CoA from pyruvic acid in the mitochondria. (3 Marks)

ii. Describe any FIVE steps of citric acid cycle. (5 Marks)

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